

MAST32001 Computational Statistics - Lecture 6

Bootstrap and jackknife for iid data

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Today

1. What problem does the bootstrap solve?
2. Parametric and nonparametric bootstrap
3. The empirical distribution and sampling with replacement
4. Bootstrap standard errors, bias, and confidence intervals
5. Bootstrap assumptions and failure cases
6. The jackknife
7. Interactive bootstrap investigation in R

The basic problem

Sampling distributions

Let

$$X_1, \dots, X_n \stackrel{iid}{\sim} F$$

and let

$$\hat{\theta} = T(X_1, \dots, X_n)$$

estimate a population quantity

$$\theta = t(F).$$

For inference we usually need the sampling distribution of

$$\hat{\theta} - \theta$$

or of a standardized version.

But the distribution F is unknown.

Classical analytic route

For some statistics we know approximate formulas.

Example:

$$\bar{X} \approx N\left(\mu, \frac{\sigma^2}{n}\right)$$

for large n , so

$$\text{se}(\bar{X}) \approx \frac{s}{\sqrt{n}}.$$

But for many statistics, such as medians, quantiles, ratios, correlations, or complex estimators, analytic standard errors can be difficult.

The bootstrap replaces some analytic derivations by computation.

Bootstrap idea

The bootstrap approximates the unknown population distribution F by an estimated distribution $\hat{F}$.

Then it simulates new samples from $\hat{F}$.

The basic analogy is

$$F \longrightarrow X_1, \dots, X_n \longrightarrow \hat{\theta}$$

and

$$\hat{F} \longrightarrow X_1^*, \dots, X_n^* \longrightarrow \hat{\theta}^*.$$

If $\hat{F}$ is a good approximation to F , then the conditional distribution of $\hat{\theta}^*$ given the data can approximate the sampling distribution of $\hat{\theta}$.

Bootstrap as plug-in principle

The bootstrap is a computational plug-in method.

Instead of deriving

$$\mathcal{L}_F\{\hat{\theta} - \theta\},$$

we approximate it by

$$\mathcal{L}_{\hat{F}}\{\hat{\theta}^* - \hat{\theta} \mid X_1, \dots, X_n\}.$$

The true distribution F and true parameter $\theta = t(F)$ are replaced by

$$\hat{F} \quad \text{and} \quad \hat{\theta} = t(\hat{F}).$$

This is why the empirical distribution is central.

Parametric and nonparametric models

What is a statistical model?

A statistical model is a collection of possible distributions for the data:

$$\mathcal{P} = \{P : \text{possible data-generating distributions}\}.$$

The model describes which data-generating distributions we regard as possible.

Two broad classes are:

- ▶ **parametric:** the distributions are indexed by a fixed finite-dimensional parameter;
- ▶ **nonparametric:** the distribution is not restricted to such a fixed finite-dimensional family.

Importantly,

nonparametric $\neq$ assumption-free.

For example, we may still assume

$$X_1, \dots, X_n \stackrel{iid}{\sim} F.$$

A parametric model

Suppose

$$X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma^2).$$

Then

$$\mathcal{P} = \{N(\mu, \sigma^2) : \mu \in \mathbb{R}, \sigma > 0\}.$$

Every possible distribution is described by the two-dimensional parameter

$$\theta = (\mu, \sigma).$$

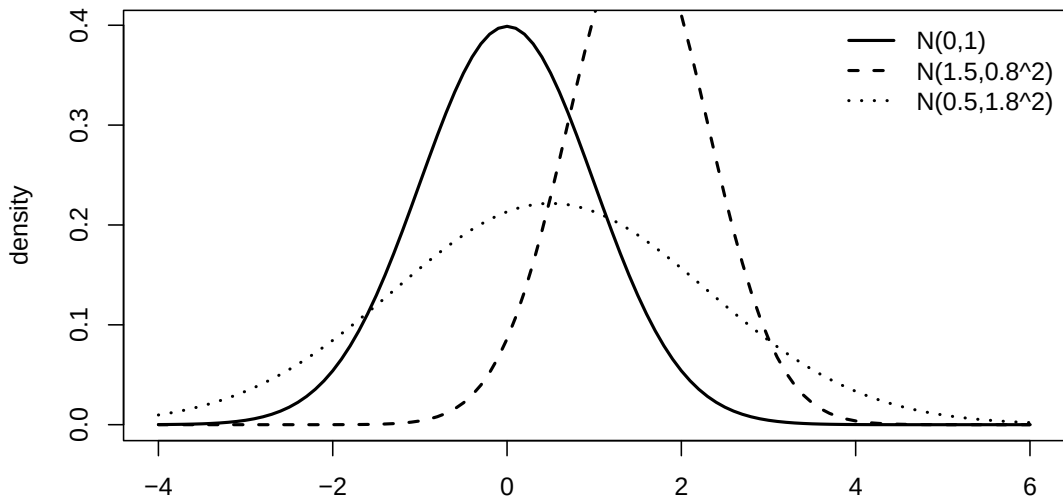
Although $\mathcal{P}$ contains infinitely many distributions, its dimension is fixed and finite.

After observing the data we estimate the parameter, for example by

$$\hat{\mu} = \bar{X}, \quad \hat{\sigma}^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2.$$

A parametric family: picture

One family, different parameter values



A nonparametric model

Now suppose only

$$X_1, \dots, X_n \stackrel{iid}{\sim} F,$$

where F may be any distribution in a broad class.

The unknown object is now the **whole distribution** F .

There is no fixed finite-dimensional vector

$$(\theta_1, \dots, \theta_d)$$

that describes all possible F .

We can still be interested in a finite-dimensional feature of F , for example

$$\mu(F) = \int x dF(x)$$

or

$$m(F) = F^{-1}(0.5),$$

without assuming that F belongs to a particular parametric family.

Nonparametric does not mean no structure

The nonparametric iid model still says something substantial:

$$X_1, \dots, X_n \stackrel{iid}{\sim} F.$$

Thus we assume:

- ▶ the observations have the same distribution;
- ▶ the observations are independent;
- ▶ one common distribution F generates all observations.

What we do **not** assume is a particular finite-dimensional form such as

$$F = N(\mu, \sigma^2).$$

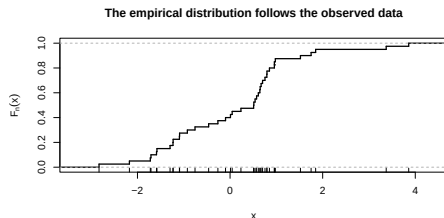
This distinction becomes crucial for deciding how to bootstrap.

Estimating the whole distribution

In the iid nonparametric model, the natural estimator of F is the empirical distribution function

$$\hat{F}_n(x) = \frac{1}{n} \sum_{i=1}^n I(X_i \leq x).$$

It puts probability $1/n$ on each observation.



This $\hat{F}_n$ will be the model from which the nonparametric bootstrap samples.

Parametric versus nonparametric

	Parametric	Nonparametric
Unknown object	$\theta \in \mathbb{R}^d$	F
Dimension	fixed, finite	not fixed finite-dimensional
Example	$N(\mu, \sigma^2)$	general F
Estimate of distribution	$F_{\hat{\theta}}$	$\hat{F}_n$
Bootstrap from	$F_{\hat{\theta}}$	$\hat{F}_n$

The bootstrap principle is the same in both cases:

estimate the data-generating distribution, then simulate from it.

Parametric and nonparametric bootstrap

Parametric bootstrap

Suppose we assume a parametric family

$$F_{\theta}, \quad \theta \in \Theta.$$

Estimate the parameter:

$$\hat{\theta}.$$

Then simulate bootstrap samples from the fitted model:

$$X_1^*, \dots, X_n^* \stackrel{iid}{\sim} F_{\hat{\theta}}.$$

For each sample compute the statistic

$$\hat{\theta}^* = T(X_1^*, \dots, X_n^*).$$

Parametric bootstrap algorithm

1. Fit the parametric model $F_{\hat{\theta}}$.
2. For $b = 1, \dots, B$:
 - ▶ simulate

$$X_1^{*(b)}, \dots, X_n^{*(b)} \sim F_{\hat{\theta}};$$

- ▶ compute

$$\hat{\theta}^{*(b)} = T(X^{*(b)}).$$

3. Use

$$\hat{\theta}^{*(1)}, \dots, \hat{\theta}^{*(B)}$$

to estimate standard errors, bias, intervals, or test distributions.

Good when the parametric model is plausible.

Bad when the fitted model is badly misspecified.

Nonparametric model

In the iid nonparametric setting, the unknown distribution F is unrestricted.

The natural estimator of F is the empirical distribution function

$$\hat{F}_n(x) = \frac{1}{n} \sum_{i=1}^n I(X_i \leq x).$$

This distribution puts mass $1/n$ on each observed value.

The nonparametric bootstrap simulates from $\hat{F}_n$.

Empirical distribution

If the observed data are

$$x_1, \dots, x_n,$$

then

$$\hat{F}_n$$

is the discrete distribution

$$P_{\hat{F}_n}(X^* = x_i) = \frac{1}{n}, \quad i = 1, \dots, n.$$

Thus drawing from $\hat{F}_n$ is exactly drawing from the observed values with equal probability.

This is the link between the empirical distribution and sampling with replacement.

Sampling with replacement

A bootstrap sample is

$$X_1^*, \dots, X_n^* \stackrel{iid}{\sim} \hat{F}_n.$$

Equivalently, draw indices

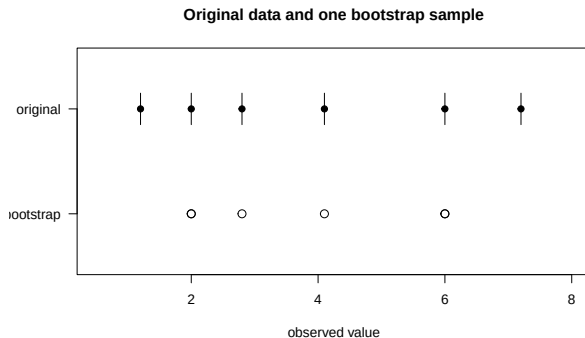
$$I_1, \dots, I_n \stackrel{iid}{\sim} \text{Unif}\{1, \dots, n\}$$

and set

$$X_j^* = X_{I_j}.$$

The same original observation can appear multiple times; some observations may not appear at all.

One bootstrap sample



The bootstrap sample contains only values that were observed.

Some observations are repeated; others are absent.

That is exactly what sampling iid from $\hat{F}_n$ means.

Nonparametric bootstrap algorithm

Given data $x = (x_1, \dots, x_n)$:

1. For $b = 1, \dots, B$:

▶ sample

$$x_1^{*(b)}, \dots, x_n^{*(b)}$$

with replacement from x ;

▶ compute

$$\hat{\theta}^{*(b)} = T(x^{*(b)}).$$

2. Use the empirical distribution of the bootstrap estimates.

The bootstrap distribution is conditional on the observed data.

R: one bootstrap sample

```
x <- c(3.2, 4.1, 5.5, 6.0, 9.4)
set.seed(20260812)
x_star <- sample(x, size = length(x), replace = TRUE)
x
```

```
## [1] 3.2 4.1 5.5 6.0 9.4
```

```
x_star
```

```
## [1] 9.4 4.1 6.0 4.1 5.5
```

```
table(x_star)
```

```
## x_star
```

```
## 4.1 5.5 6 9.4
```

```
## 2 1 1 1
```

The bootstrap sample has the same size as the original sample, but not necessarily the same distinct observations.

How many distinct observations?

For one observation x_i ,

$$P(x_i \text{ is not selected}) = \left(1 - \frac{1}{n}\right)^n.$$

As $n \rightarrow \infty$,

$$\left(1 - \frac{1}{n}\right)^n \rightarrow e^{-1}.$$

So the expected fraction of original observations that appear at least once is approximately

$$1 - e^{-1} \approx 0.632.$$

A bootstrap sample usually contains repeated observations.

Demonstration of the 0.632 phenomenon

```
set.seed(20260812)

n <- 100
B <- 5000

distinct_fraction <- replicate(B, {
  ind <- sample.int(n, size = n, replace = TRUE)
  length(unique(ind)) / n
})

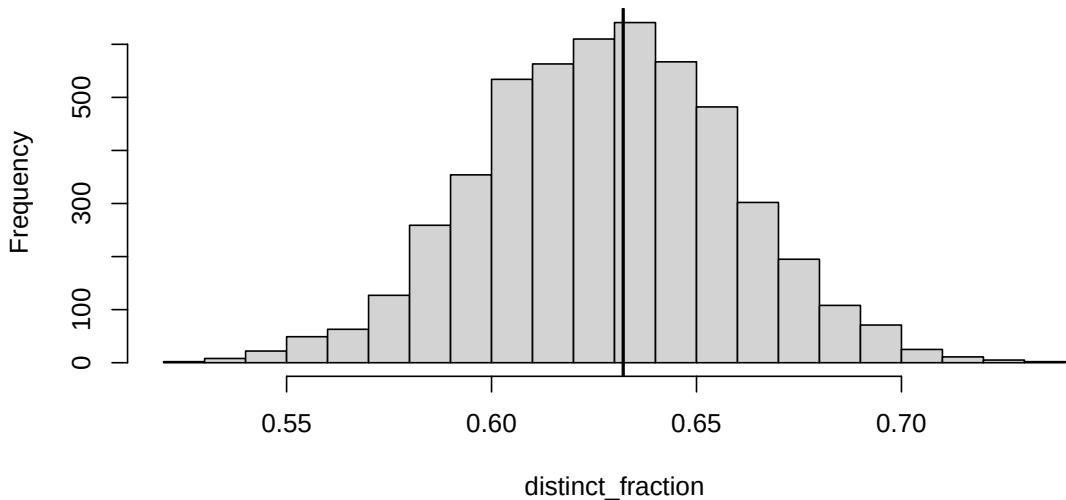
c(mean = mean(distinct_fraction), theoretical = 1 - exp(-1))

##          mean theoretical
## 0.6331320  0.6321206

hist(distinct_fraction, breaks = 30,
     main = "Fraction of distinct observations in bootstrap samples")
abline(v = 1 - exp(-1), lwd = 2)
```

Demonstration of the 0.632 phenomenon

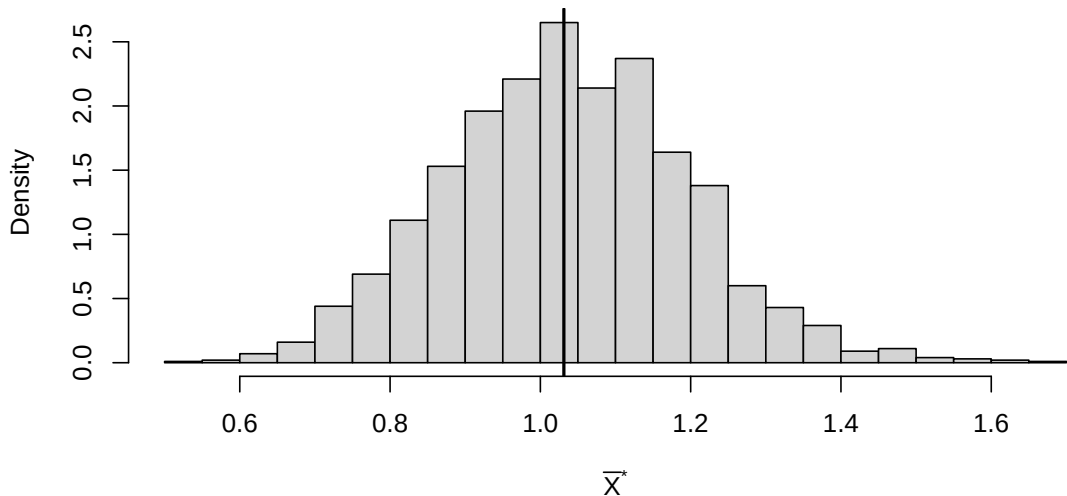
Fraction of distinct observations in bootstrap samples



Bootstrap for estimation

Many resamples give a bootstrap distribution

2000 bootstrap estimates



How many bootstrap replicates?

Suppose we generate $\hat{\theta}^{*(1)}, \dots, \hat{\theta}^{*(B)}$.

The number B controls the **Monte Carlo accuracy** with which we approximate the bootstrap distribution.

It does **not** increase the information in the original sample.

For example, for a bootstrap probability

$$\hat{p}_B = \frac{1}{B} \sum_{b=1}^B I(T_b^* \leq t),$$

conditional on the observed data,

$$\text{SE}_{MC}(\hat{p}_B) = \sqrt{\frac{p^*(1-p^*)}{B}}.$$

Hence the Monte Carlo error decreases at rate

$$O(B^{-1/2}).$$

So reducing Monte Carlo error by a factor of 10 requires roughly 100 times as many bootstrap replicates.

Choosing B in practice

There is no universal best value of B . It depends on what we want to estimate.

A useful practical guide is:

- ▶ exploratory plots and teaching examples: $B \approx 1000$ may be enough;
- ▶ bootstrap standard errors: a few thousand replicates are often adequate;
- ▶ confidence-interval quantiles: use more, for example $B = 5000$ – 10000 ;
- ▶ small tail probabilities or bootstrap tests: potentially much larger B .

For a Monte Carlo bootstrap test with

$$p_{\text{boot}} = \frac{1 + \sum_{b=1}^B I(T_b^* \geq T_{\text{obs}})}{B + 1},$$

the p-value resolution is approximately

$$\frac{1}{B + 1}.$$

Thus $B = 999$ gives resolution 0.001, whereas $B = 9999$ gives resolution 0.0001.

Choose B so that Monte Carlo error is negligible for the purpose at hand.

Increasing B reduces simulation error; increasing n addresses statistical sampling uncertainty.

Bootstrap standard error

Let

$$\hat{\theta}^{*(1)}, \dots, \hat{\theta}^{*(B)}$$

be bootstrap estimates.

The bootstrap estimate of standard error is

$$\widehat{\text{se}}_{\text{boot}}(\hat{\theta}) = \sqrt{\frac{1}{B-1} \sum_{b=1}^B \left(\hat{\theta}^{*(b)} - \bar{\hat{\theta}}^* \right)^2},$$

where

$$\bar{\hat{\theta}}^* = \frac{1}{B} \sum_{b=1}^B \hat{\theta}^{*(b)}.$$

Bootstrap bias estimate

The bias of $\hat{\theta}$ is

$$\text{bias}(\hat{\theta}) = E_F(\hat{\theta}) - \theta.$$

The bootstrap estimate is

$$\widehat{\text{bias}}_{\text{boot}} = \bar{\hat{\theta}}^* - \hat{\theta}.$$

A bias-corrected estimator is

$$\hat{\theta}_{\text{bc}} = \hat{\theta} - \widehat{\text{bias}}_{\text{boot}} = 2\hat{\theta} - \bar{\hat{\theta}}^*.$$

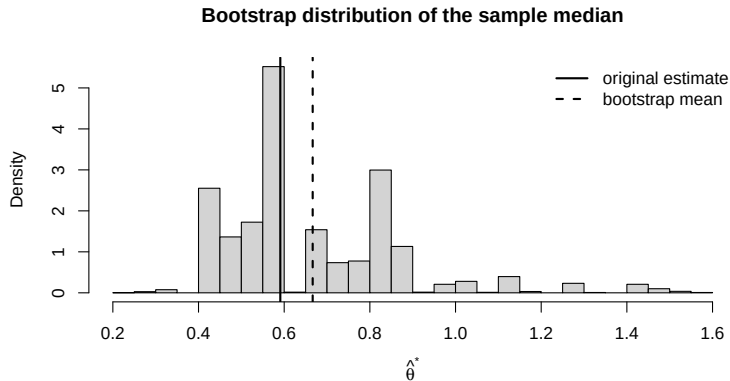
Bias correction can increase variance, so it should not be used automatically.

Example: standard error of a median

```
set.seed(20260812)
x <- rexp(40, rate = 1)
theta_hat <- median(x)
B <- 5000
theta_star <- replicate(B, {
  xs <- sample(x, size = length(x), replace = TRUE)
  median(xs)
})
c(
  median = theta_hat,
  boot_se = sd(theta_star),
  boot_bias = mean(theta_star) - theta_hat
)
```

```
##      median    boot_se boot_bias
## 0.59092716 0.20635426 0.07559794
```

Bootstrap distribution



The bootstrap distribution approximates the sampling distribution of the statistic around the observed estimate.

Bootstrap confidence intervals

Several bootstrap confidence intervals exist.

Common introductory versions:

Normal bootstrap interval

$$\hat{\theta} \pm z_{1-\alpha/2} \hat{\text{se}}_{\text{boot}}.$$

Percentile interval

$$\left[q_{\alpha/2}(\hat{\theta}^*), q_{1-\alpha/2}(\hat{\theta}^*) \right].$$

Basic bootstrap interval

$$\left[2\hat{\theta} - q_{1-\alpha/2}(\hat{\theta}^*), 2\hat{\theta} - q_{\alpha/2}(\hat{\theta}^*) \right].$$

Percentile interval example

```
alpha <- 0.05
ci_percentile <- quantile(theta_star, probs = c(alpha / 2, 1 - alpha / 2))
ci_normal <- theta_hat +
  c(-1, 1) * qnorm(1 - alpha / 2) * sd(theta_star)
rbind(
  percentile = ci_percentile,
  normal = ci_normal
)
```

```
##           2.5%      97.5%
## percentile 0.4203644 1.2639091
## normal     0.1864802 0.9953741
```

The percentile interval is easy to explain, but it is not automatically accurate.

Bootstrap intervals depend on assumptions and on the statistic.

Bootstrap assumptions

Bootstrap is not assumption-free

Bootstrap methods are sometimes introduced as if they require no assumptions.

That is misleading.

The bootstrap replaces unavailable mathematical derivations with simulation from an estimated data-generating mechanism.

Therefore validity requires that the estimated mechanism captures the relevant features of the true sampling process.

A useful question is:

What data-generating process is my resampling scheme pretending to imitate?

Assumptions for the simple iid bootstrap

The ordinary nonparametric bootstrap assumes, at least approximately:

- ▶ observations are iid or exchangeable in the relevant way;
- ▶ the empirical distribution is a reasonable approximation to the population distribution;
- ▶ the statistic is sufficiently regular or smooth;
- ▶ the sample size is large enough for the plug-in approximation to be useful;
- ▶ the sampling design is represented by the resampling scheme.

These assumptions should be discussed explicitly.

Smoothness of the statistic

The bootstrap works best for statistics that change smoothly when the distribution changes.

Examples usually suitable:

- ▶ mean;
- ▶ variance;
- ▶ regression coefficients under regular conditions;
- ▶ smooth functions of moments.

Examples requiring care:

- ▶ maximum;
- ▶ sample extremes;
- ▶ hard-thresholded model selection;
- ▶ nonsmooth statistics at boundary cases;
- ▶ nearest-neighbour rules.

The sample median is less smooth than the mean but often bootstrappable under continuity and positive density at the median.

Sample size and discreteness

The nonparametric bootstrap samples from a discrete distribution with only n support points.

Consequences:

- ▶ extreme values cannot be generated beyond the observed range;
- ▶ rare categories may disappear or be overrepresented;
- ▶ tails are represented only by observed tail points;
- ▶ bootstrap distributions can be discrete, especially for small n .

This is not a bug. It follows from using $\hat{F}_n$.

Dependence and design

The iid bootstrap is inappropriate when observations are not iid.

Examples:

- ▶ time series;
- ▶ spatial data;
- ▶ clustered data;
- ▶ repeated measurements;
- ▶ survey data with unequal weights;
- ▶ matched pairs.

In such cases we need a resampling scheme that preserves the relevant dependence or sampling structure.

This motivates block bootstrap, cluster bootstrap, pairs bootstrap, and other variants.

When the bootstrap can fail

Examples where a naive bootstrap can fail or be misleading:

- ▶ estimating the endpoint of a distribution, such as $F = U(0, \theta)$ and $\hat{\theta} = \max X_i$;
- ▶ heavy-tailed data where variance is infinite or poorly estimated;
- ▶ very small sample sizes;
- ▶ dependent data resampled as iid;
- ▶ statistics after model selection;
- ▶ parameters on the boundary;
- ▶ rare-event probabilities with no rare events observed.

The issue is usually not R code. It is a mismatch between the statistical problem and the resampling scheme.

Example: maximum of a uniform distribution

Let

$$X_1, \dots, X_n \sim U(0, \theta)$$

and estimate

$$\theta$$

by

$$\hat{\theta} = \max_i X_i.$$

The nonparametric bootstrap samples only from observed values.

Therefore

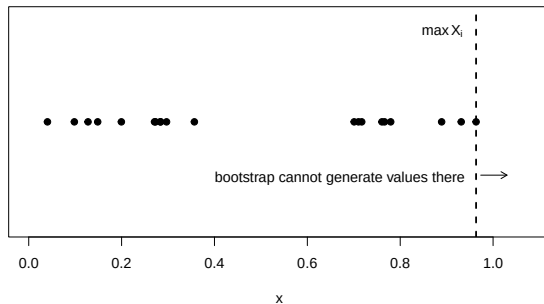
$$\max_i X_i^* \leq \max_i X_i$$

always.

The bootstrap cannot generate samples with maxima above the observed maximum, even though real future samples could exceed it.

Why the maximum is different

The empirical distribution has no mass beyond the observed maximum



For a mean, the empirical distribution can often mimic the relevant local variation.

For an endpoint, the missing part beyond the observed maximum is exactly the part needed for the sampling distribution.

R: bootstrap of the maximum

```
set.seed(20260812)
n <- 30
x <- runif(n, 0, 1)
theta_hat <- max(x)

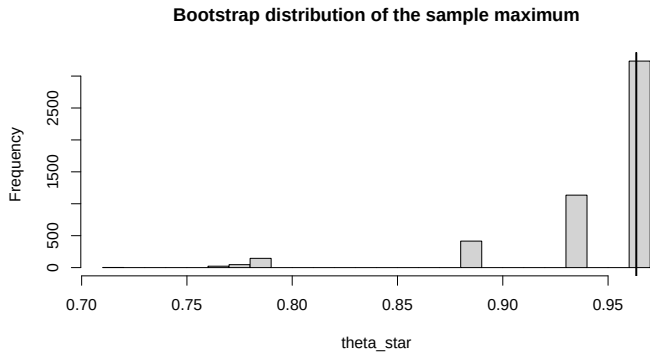
B <- 5000
theta_star <- replicate(B, max(sample(x, n, replace = TRUE)))

c(original_max = theta_hat,
  max_bootstrap_max = max(theta_star),
  mean_bootstrap_max = mean(theta_star))

##          original_max  max_bootstrap_max  mean_bootstrap_max
##          0.9634004      0.9634004      0.9420549

hist(theta_star, breaks = 30,
      main = "Bootstrap distribution of the sample maximum")
abline(v = theta_hat, lwd = 2)
```

R: bootstrap of the maximum



This is a structural failure of the naive nonparametric bootstrap for this parameter.

Interactive part

R TASK: bootstrap assumptions in action

Work with your neighbour.

Use the same data set:

```
set.seed(20260812)
x <- runif(30, 0, 1)
B <- 3000
```

Compare bootstrap distributions for two statistics:

$$T_1 = \bar{X}, \quad T_2 = \max X_i.$$

Tasks:

1. Bootstrap both statistics using sampling with replacement.
2. Plot both bootstrap distributions.
3. For which statistic does the ordinary bootstrap seem more plausible?
4. What feature of the maximum causes the problem?

R TASK: one possible solution

```
set.seed(20260812)

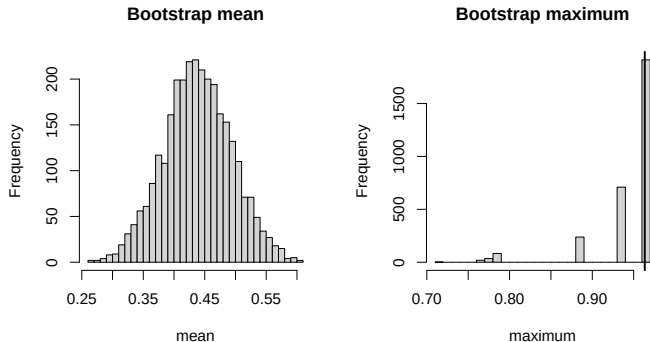
x <- runif(30, 0, 1)
B <- 3000

boot_mean <- replicate(B, mean(sample(x, length(x), replace = TRUE)))
boot_max <- replicate(B, max(sample(x, length(x), replace = TRUE)))

c(
  mean_original = mean(x),
  mean_boot_se = sd(boot_mean),
  max_original = max(x),
  max_largest_boot = max(boot_max)
)
```

##	mean_original	mean_boot_se	max_original	max_largest_boot
##	0.43962768	0.05504198	0.96340044	0.96340044

R TASK: plots



The mean is a smooth statistic and the bootstrap distribution looks plausible.

The maximum is constrained by the observed maximum, so the bootstrap cannot represent possible larger future maxima.

Jackknife

The jackknife

The jackknife is an older resampling method based on leaving out one observation at a time.

Given data

$$x = (x_1, \dots, x_n),$$

define the leave-one-out sample

$$x_{(-i)} = (x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n).$$

The jackknife replicate is

$$\hat{\theta}_{(-i)} = T(x_{(-i)}).$$

There are exactly n jackknife replicates.

Jackknife standard error

Let

$$\bar{\theta}_{(\cdot)} = \frac{1}{n} \sum_{i=1}^n \hat{\theta}_{(-i)}.$$

The jackknife standard error estimate is

$$\hat{\text{se}}_{\text{jack}} = \sqrt{\frac{n-1}{n} \sum_{i=1}^n \left(\hat{\theta}_{(-i)} - \bar{\theta}_{(\cdot)} \right)^2}.$$

The factor $(n-1)/n$ accounts for the strong dependence between leave-one-out samples.

Jackknife bias estimate

The jackknife estimate of bias is

$$\widehat{\text{bias}}_{\text{jack}} = (n - 1) \left(\bar{\theta}_{(\cdot)} - \hat{\theta} \right).$$

The corresponding bias-corrected estimate is

$$\hat{\theta}_{\text{jack}} = \hat{\theta} - \widehat{\text{bias}}_{\text{jack}} = n\hat{\theta} - (n - 1)\bar{\theta}_{(\cdot)}.$$

This was one of the original motivations for the jackknife.

Jackknife works best for smooth statistics

The jackknife is most reliable for statistics that respond smoothly to deleting one observation.

Good examples:

- ▶ mean;
- ▶ smooth functions of means;
- ▶ regression coefficients under regular conditions.

Problematic examples:

- ▶ median in some small or discrete samples;
- ▶ maximum;
- ▶ nonsmooth selection rules.

This is similar to, but stronger than, the smoothness issue for the bootstrap.

R: jackknife standard error

```
jackknife <- function(x, statistic) {  
  n <- length(x)  
  sapply(seq_len(n), function(i) statistic(x[-i]))  
}  
x <- rexp(40)  
jack_mean <- jackknife(x, mean)  
jack_median <- jackknife(x, median)  
  
jack_se_mean <- sqrt((length(x) - 1) / length(x) *  
                      sum((jack_mean - mean(jack_mean))^2))  
jack_se_median <- sqrt((length(x) - 1) / length(x) *  
                       sum((jack_median - mean(jack_median))^2))  
c(mean = jack_se_mean, median = jack_se_median)  
  
##      mean      median  
## 0.2436593 0.6162409
```

Bootstrap versus jackknife

	Bootstrap	Jackknife
Samples	B resamples	n leave-one-out samples
Random?	usually yes	no
Main use	distribution, SE, CI	SE, bias, diagnostics
Works best	regular statistics	smooth statistics
Output	full simulated distribution	leave-one-out sensitivity

The bootstrap is more flexible.

The jackknife is deterministic and can reveal influential observations.

Summary

- ▶ The nonparametric bootstrap samples from the empirical distribution.
- ▶ Sampling from the empirical distribution is sampling with replacement from the observed data.
- ▶ Bootstrap standard errors, bias estimates, and confidence intervals are based on the bootstrap distribution of the statistic.
- ▶ Bootstrap is not assumption-free.
- ▶ The ordinary iid bootstrap requires the iid sampling structure to be plausible.
- ▶ Smooth statistics are easier to bootstrap than nonsmooth statistics.
- ▶ The jackknife uses leave-one-out samples and is useful for standard errors, bias, and diagnostics.